

Numerical Road to Quantum dots

I will use a 5-point stencil for the $\frac{d^2}{dx^2}$ operator:

Having:

$$\begin{aligned} \bullet f(x_k + h) &= f(x_{k+1}) \approx f(x_k) + hf'(x_k) + \frac{h^2}{2}f''(x_k) + \dots & \bullet f(x_k - h) &= f(x_{k-1}) \approx f(x_k) - hf'(x_k) + \frac{h^2}{2}f''(x_k) - \dots \\ \bullet f(x_k + 2h) &= f(x_{k+2}) \approx f(x_k) + 2hf'(x_k) + 2h^2f''(x_k) + \dots & \bullet f(x_k - 2h) &= f(x_{k-2}) \approx f(x_k) - 2hf'(x_k) + 2h^2f''(x_k) - \dots \\ \bullet f(x_k + 3h) &= f(x_{k+3}) \approx f(x_k) + 3hf'(x_k) + \frac{9}{2}h^2f''(x_k) + \dots & \bullet f(x_k - 3h) &= f(x_{k-3}) \approx f(x_k) - 3hf'(x_k) + \frac{9}{2}h^2f''(x_k) - \dots \\ \bullet f(x_k + 4h) &= f(x_{k+4}) \approx f(x_k) + 4hf'(x_k) + 8h^2f''(x_k) + \dots & \bullet f(x_k - 4h) &= f(x_{k-4}) \approx f(x_k) - 4hf'(x_k) + 8h^2f''(x_k) - \dots \\ \bullet f(x_k + 5h) &= f(x_{k+5}) \approx f(x_k) + 5hf'(x_k) + \frac{25}{2}h^2f''(x_k) + \dots & \bullet f(x_k - 5h) &= f(x_{k-5}) \approx f(x_k) - 5hf'(x_k) + \frac{25}{2}h^2f''(x_k) - \dots \end{aligned}$$

Adding them to get rid of $f', f^{(3)}, f^{(4)}$.

$$\rightarrow -154f(x_{k+1}) + 214f(x_{k+2}) - 156f(x_{k+3}) + 61f(x_{k+4}) - 10f(x_{k+5}) = -45f(x_k) + (-77 + 428 - 702 + 488 - 125)h^2f''(x_k)$$

$$\rightarrow f''(x_k) = \frac{45f(x_k) - 154f(x_{k+1}) + 214f(x_{k+2}) - 156f(x_{k+3}) + 61f(x_{k+4}) - 10f(x_{k+5})}{12h^2}$$

$$\rightarrow 11f(x_{k-1}) + 6f(x_{k+1}) + 4f(x_{k+2}) - f(x_{k+3}) = 20f(x_k) + \left(\frac{11}{2} + 3 + 8 - \frac{9}{2}\right)h^2f''(x_k)$$

$$\rightarrow f''(x_k) = \frac{11f(x_{k-1}) - 20f(x_k) + 6f(x_{k+1}) + 4f(x_{k+2}) - f(x_{k+3})}{12h^2}$$

$$\rightarrow -f(x_{k-2}) + 16f(x_{k-1}) + 16f(x_{k+1}) - f(x_{k+2}) = 30f(x_k) + (-2 + 8 + 8 - 2)h^2f''(x_k)$$

$$\rightarrow f''(x_k) = \frac{-f(x_{k-2}) + 16f(x_{k-1}) - 30f(x_k) + 16f(x_{k+1}) - f(x_{k+2})}{12h^2}$$

$$\rightarrow -f(x_{k-3}) + 4f(x_{k-2}) + 6f(x_{k-1}) + 11f(x_{k+1}) = 20f(x_k) + \left(-\frac{9}{2} + 8 + 3 + \frac{11}{2}\right)h^2f''(x_k)$$

$$\rightarrow f''(x_k) = \frac{-f(x_{k-3}) + 4f(x_{k-2}) + 6f(x_{k-1}) - 20f(x_k) + 11f(x_{k+1})}{12h^2}$$

$$\rightarrow -10f(x_{k+3}) + 61f(x_{k+4}) - 156f(x_{k+5}) + 214f(x_{k+2}) - 154f(x_{k+1}) = -45f(x_k) + (-77 + 428 - 702 + 488 - 125)h^2f''(x_k)$$

$$\rightarrow f''(x_k) = \frac{-10f(x_{k+3}) + 61f(x_{k+4}) - 156f(x_{k+5}) + 214f(x_{k+2}) - 154f(x_{k+1}) + 45f(x_k)}{12h^2}$$

Summarized:

$$* f''(x_0) = \frac{45f(x_0) - 154f(x_1) + 214f(x_2) - 156f(x_3) + 61f(x_4) - 10f(x_5)}{12h^2}$$

$$* f''(x_1) = \frac{11f(x_0) - 20f(x_1) + 6f(x_2) + 4f(x_3) - f(x_4)}{12h^2}$$

$$* f''(x_k) = \frac{-f(x_{k-2}) + 16f(x_{k-1}) - 30f(x_k) + 16f(x_{k+1}) - f(x_{k+2})}{12h^2}$$

$$* f''(x_{n-1}) = \frac{-f(x_{n-3}) + 4f(x_{n-2}) + 6f(x_{n-1}) - 20f(x_n) + 11f(x_{n+1})}{12h^2}$$

$$* f''(x_n) = \frac{-10f(x_{n-5}) + 61f(x_{n-4}) - 156f(x_{n-3}) + 214f(x_{n-2}) - 154f(x_{n-1}) + 45f(x_n)}{12h^2}$$

Assuming Dirichlet boundary conditions: $\Psi(x_{\min}) = \Psi(x_{\max}) = 0$, we can just use the 4th-order central stencil

$$* \Psi''(x_k) = \frac{-\Psi(x_{k-2}) + 16\Psi(x_{k-1}) - 30\Psi(x_k) + 16\Psi(x_{k+1}) - \Psi(x_{k+2}))}{12dx^2}$$

for all the points in the grid. This decision makes sure that our Hamiltonian is Hermitian $H = H^\dagger$

But notice for $k=1$ (first interior point):

$$\Psi''(x_1) = \frac{-\Psi_1 + 16\Psi_0 - 30\Psi_1 + 16\Psi_2 - \Psi_3}{12dx^2}$$

* And notice the subindex i in Ψ_i refers to the value of the eigenfunction at that grid point.
Not to the energy level

And two of these five values are not interior unknowns:

• $\Psi_0 = \Psi(x_0) = 0$ from the imposed Dirichlet boundary condition.

• $\Psi_{-1} = \Psi(x_0 - dx)$ → it is not a boundary value and is not zero. Let's find what it is
→ Notice that if we add the two Taylor expansions about x_0 :

$$(1) \quad \Psi(x_0 + dx) + \Psi(x_0 - dx) = 2 \sum_{j=0}^{\infty} \frac{\Psi^{(2j)}(x_0)}{(2j)!} (dx)^{2j}$$

* I originally made a mistake by assuming $\Psi_{-1} = 0$

Every odd derivative cancels in this sum and only the even ones survive. On the next page I show that the $j=0$ and $j=1$ terms both vanish so the sum starts at $j=2$ and

$$\Psi(x_0 - dx) \approx -\Psi(x_0 + dx)$$

$$\Psi_{-1} \approx -\Psi_1$$

* Showing which even derivatives vanish

→ Writing the time-independent Schrödinger equation as:

$$\Psi'' = W\Psi, \quad W(x) = \frac{2m}{\hbar^2} (V(x) - E)$$

• for $j=0$: $\Psi(x_0)=0$ → This is the Dirichlet condition itself.

• for $j=1$: $\Psi''(x_0) = W(x_0)\Psi(x_0) = W(x_0) \cdot 0 = 0$

• for $j=2$:

$$\rightarrow \Psi''' = (W\Psi)' = W'\Psi + W\Psi'$$

$$\rightarrow \Psi'''' = W''\Psi + 2W'\Psi' + W\Psi'' = \cancel{W''\Psi} + 2W'\Psi' + \cancel{W\Psi''} = 2W'\Psi'$$

$\underbrace{W\Psi}_{\text{vanish at } x_0}$

$$\Rightarrow \Psi''''(x_0) = 2W'(x_0)\Psi'(x_0) = \frac{4m}{\hbar^2} V'(x_0)\Psi'(x_0)$$

So, truncating the sum from Eq. 1:

$$\Psi(x_0+dx) + \Psi(x_0-dx) = \frac{\Psi''''(x_0)}{12} (dx)^4 + O(dx^6)$$

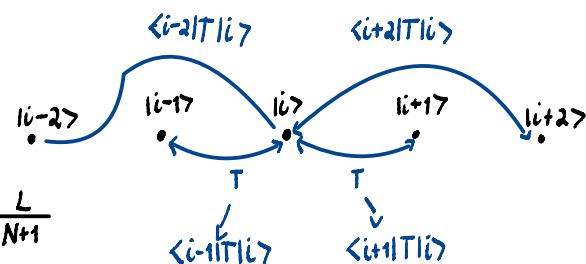
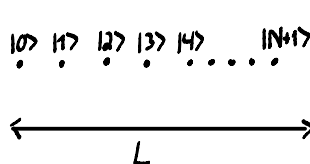
Thus,

$$(2) \cdot \Psi_{-1} = -\Psi_1 + \frac{\Psi''''(x_0)}{12} dx^4 + O(dx^6) \approx -\Psi_1$$

* Implemented in the code by adding $\frac{1}{12dx^2}$ to the first and last entry of the diagonal

→ 1-dimensional

$$H = T + V = \frac{p^2}{2m} + V = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V$$



$$\Rightarrow \langle i | H | i \rangle = \langle i | T | i \rangle + \langle i | V | i \rangle$$

$$= -\frac{\hbar^2}{2m} \left(\frac{-\langle i | i-2 \rangle + 16 \langle i | i-1 \rangle - 30 \langle i | i \rangle + 16 \langle i | i+1 \rangle - \langle i | i+2 \rangle}{12 (dx)^2} \right) + V(x_i) \delta_{i,i}$$

$$= -\frac{\hbar^2}{24m(dx)^2} (-\delta_{i,i-2} + 16 \delta_{i,i-1} - 30 \delta_{i,i} + 16 \delta_{i,i+1} - \delta_{i,i+2}) + V(x_i) \delta_{i,i}$$

Thus, the discretized second derivative entering the kinetic term is

$$\Rightarrow \Psi''_i = \frac{-\Psi_{i-2} + 16\Psi_{i-1} - 30\Psi_i + 16\Psi_{i+1} - \Psi_{i+2}}{12 dx^2}$$

and substituting Eq. 2 for $i=1, N$

$$\Rightarrow \Psi''_1 = \frac{(-30+1)\Psi_1 + 16\Psi_2 - \Psi_3}{12 dx^2}$$

$$\Rightarrow \Psi''_N = \frac{(-30+1)\Psi_N + 16\Psi_{N-1} - \Psi_{N-2}}{12 dx^2}$$

And therefore, the total Hamiltonian including the frontier correction is:

$$H_{i,i} = -\frac{\hbar^2}{24m(dx)^2} (-\delta_{i,i-2} + 16\delta_{i,i-1} - 30\delta_{i,i} + 16\delta_{i,i+1} - \delta_{i,i+2}) + V(x_i)\delta_{i,i} \quad \text{for } 2 \leq i \leq N-1$$

and specifically for the first and last ($i=1, N$) interior points:

* Quick recall: $\Psi_0 = \Psi(x_0) = 0$ and $\Psi_{N+1} = \Psi(x_{N+1}) = 0$ by Dirichlet. The remaining "out of domain" amplitudes $\Psi_{-1} = \Psi(x_{-1})$ and $\Psi_{N+2} = \Psi(x_{N+2})$ do not vanish. By Eq. (2) they fold onto the diagonal, which is where $-30 \Rightarrow -29$ comes from.

$$H_{i,1} = -\frac{\hbar^2}{24m(dx)^2} (-29\delta_{i,1} + 16\delta_{i,2} - \delta_{i,3}) + V(x_1)\delta_{i,1}, \quad H_{i,N} = -\frac{\hbar^2}{24m(dx)^2} (-29\delta_{i,N} + 16\delta_{i,N-1} - \delta_{i,N-2}) + V(x_N)\delta_{i,N}$$

I listed 9 potential functions:

1) Harmonic Oscillator

$$V(x) = \frac{1}{2}m\omega^2 x^2, \quad E_n = \hbar\omega(n + \frac{1}{2}), \quad n=0, 1, 2, 3, 4, \dots$$

2) Anharmonic Oscillator

$$V(x) = \frac{1}{2}(ax^2 + bx^4), \quad E_n \approx \hbar\omega(n + \frac{1}{2}) + \frac{3b\hbar^2}{8m^2\omega^2}(an^2 + 2n + 1), \quad n=0, 1, 2, 3, \dots$$

$$* a = m\omega^2$$

Perturbation theory:

$$H' = \frac{1}{2}bx^4, \quad x = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}^\dagger + \hat{a}), \quad \hat{a}|n\rangle = \sqrt{n}|n-1\rangle, \quad \hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

$$\text{So: } (\hat{a}^\dagger + \hat{a})^4 = \hat{a}^{\dagger 4} + \hat{a}^{\dagger 3}\hat{a} + \hat{a}^{\dagger 2}\hat{a}\hat{a}^\dagger + \hat{a}^{\dagger 2}\hat{a}^2 + \hat{a}^\dagger\hat{a}\hat{a}^{\dagger 2} + \hat{a}^\dagger\hat{a}\hat{a}^\dagger\hat{a} + \hat{a}^\dagger\hat{a}^2\hat{a}^\dagger + \hat{a}^\dagger\hat{a}^3 + \hat{a}\hat{a}^{\dagger 3} + \hat{a}\hat{a}^{\dagger 2}\hat{a}^\dagger + \hat{a}\hat{a}^\dagger\hat{a}^2 + \hat{a}\hat{a}^\dagger\hat{a}^\dagger\hat{a} + \hat{a}^2\hat{a}^{\dagger 2} + \hat{a}^2\hat{a}^\dagger\hat{a} + \hat{a}^3\hat{a}^\dagger + \hat{a}^4$$

From the expression found for $(\hat{a}^\dagger + \hat{a})^4$ we are just going to focus on the terms that contribute to $\langle n| \dots |n\rangle$ since $\langle n|n+k\rangle = 0$, for that, the terms that we want need the same amount of $\hat{a}^\dagger$ as $\hat{a}$, so:

$$\rightarrow \langle n|(\hat{a}^\dagger + \hat{a})^4|n\rangle = \langle n|(\hat{a}^{\dagger 2}\hat{a}^2 + \hat{a}^\dagger\hat{a}\hat{a}^\dagger\hat{a} + \hat{a}^\dagger\hat{a}^2\hat{a}^\dagger + \hat{a}^\dagger\hat{a}^{\dagger 2}\hat{a} + \hat{a}\hat{a}^\dagger\hat{a}^2 + \hat{a}^2\hat{a}^{\dagger 2})|n\rangle$$

$$= (\langle n|)(\langle n-1|)(\langle n-1|)(\langle n|) + (\langle n|)(\langle n|)(\langle n|)(\langle n|) + (\langle n+1|)(\langle n+1|)(\langle n|)(\langle n|) + (\langle n|)(\langle n|)(\langle n+1|)(\langle n+1|) + (\langle n+1|)(\langle n+1|)(\langle n+1|)(\langle n+1|) + (\langle n+1|)(\langle n+2|)(\langle n+2|)(\langle n+1|)$$

$$= n(n-1) + n^2 + (n+1)n + n(n+1) + (n+1)^2 + (n+1)(n+2)$$

$$= n^2 - n + n^2 + n^2 + n + n^2 + n + n^2 + 2n + 1 + n^2 + 3n + 2$$

$$= 6n^2 + 6n + 3$$

$$\rightarrow \langle n|\frac{1}{2}bx^4|n\rangle = \frac{b\hbar^2}{8m^2\omega^2}(6n^2 + 6n + 3) = \frac{3b\hbar^2}{8m^2\omega^2}(2n^2 + 2n + 1)$$

3) Infinite Square Well

$$V(x) = \begin{cases} \infty, & x < 0, x > L \\ 0, & 0 \leq x \leq L \end{cases}, \quad E_n = \frac{\hbar^2 \pi^2}{2mL^2} n^2, \quad n=1,2,3,\dots$$

4) Finite Square Well

$$V(x) = \begin{cases} V_0, & x < 0, x > L \\ 0, & 0 \leq x \leq L \end{cases}$$

• Solve for even states:

$$kL \tan\left(\frac{kL}{2}\right) = \kappa L$$

• Solve for odd states:

$$kL \cot\left(\frac{kL}{2}\right) = -\kappa L$$

with: $k = \frac{\sqrt{2mE}}{\hbar}, \quad \kappa = \frac{\sqrt{2m(V_0-E)}}{\hbar}$

5) Linear Potential

$V(x) = Fx$, its solution reduces to an Airy equation and so the energy eigenvalues are determined by Airy zeros (considering Dirichlet boundary conditions)

6) Soft Coulomb

$$V(x) = -\frac{Z}{\sqrt{x^2 + \epsilon^2}}, \quad \text{no closed form, it depends on } \epsilon$$

7) Quartic Single Well

$$V(x) = V_0 x^4, \quad \text{No analytic spectrum}$$

8) Discrete Delta Well

$$V(x) = -\alpha \delta(x), \quad E = -\frac{m\alpha^2}{2\hbar^2}$$

9) Quartic Double Well

$$V(x) = V_0(x^2 - a^2)^2, \quad \text{No closed form.}$$

- * Nearly degenerate ground pair
- * Tunneling